

Dehao Yuan

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[GitHub](#)

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Machine Learning Computer Vision Event Camera Point Cloud Large Language Model
python C++ CUDA numpy pytorch transformers opencv flask
transformers accelerate deepspeed trl vllm distributed training

**I can pick up new expertise very quickly,
given my strong math background, rich engineering and research experience.**

Education & Employment

Applied Researcher I, Manager.

Jun. 2, 2025 — Now

Capital One, McLean VA.

- Pre-trained foundation models using large-scale sequential credit card event data, through masked language modeling (MLM) and causal language modeling (CLM), with frameworks transformers, accelerate, and deepspeed for efficient multi-node distributed training.
- Post-trained foundation models for six financial use cases, such as first party fraud.
- Systematically research model improvement, including time embedding, numeric embedding, architecture variation e.g. hierarchical chunking models.
- Generated large-scale credit card event datasets by applying knowledge distillation to large language models, integrated into high-throughput inference pipelines using vllm. The synthetic data are augmentation of the real-world data.

Ph.D. in Computer Science.

Aug. 23, 2021 — May 22, 2025

University of Maryland, College Park (UMD).

- Thesis Title: Vector Kernel Representation Theory: Applications in Point Clouds and Event Cameras

B.Sc. in Mathematics and Economics & Data Science and Technology.

Sep. 1, 2017 — Jul. 14, 2021

Hong Kong University of Science and Technology (HKUST).

- Worked as an undergraduate research assistant on computational environmental science for three years, co-authored 7 papers.
- First-class Honor, Chern-class Scholarship (Top 1% math-major graduates).

Publication — as a Ph.D. student at UMD.

1. A Real-Time Event-Based Normal Flow Estimator.

[D. Yuan](#), C. Fermüller.

Technical Report.

[Paper](#) [Code](#)

2. Learning Normal Flow And Egomotion Directly From Events.

[D. Yuan](#), L. Burner, J. Wu, M. Liu, J. Chen, Y. Aloimonos, C. Fermüller.

International Conference on Computer Vision, [ICCV 2025](#).

[Paper](#) [Code](#)

3. Repurposing Pre-trained Video Diffusion Models for Event-based Video Interpolation.

J. Chen, B.Y. Feng, H. Cai, T. Wang, L. Burner, [D. Yuan](#), C. Fermüller, C.A. Metzler, Y. Aloimonos.

The IEEE/CVF Conference on Computer Vision and Pattern Recognition, [CVPR 2025](#).

[Paper](#) [Code](#)

4. A Linear Time and Space Local Point Cloud Geometry Encoder via Vectorized Kernel Mixture (VecKM).

[D. Yuan](#), C. Fermüller, T. Rabbani, F. Huang, Y. Aloimonos.

International Conference on Machine Learning, [ICML 2024](#).

[Paper](#) [Code](#)

5. Decodable and Sample Invariant Continuous Object Encoder.

[D. Yuan](#), F. Huang, C. Fermüller, Y. Aloimonos.

International Conference on Learning Representation, [ICLR 2024](#).

[Paper](#) [Code](#)

6. Discovering Object Attributes by Prompting Large Language Models with Perception-Action APIs.
A. Mavrogiannis, [D. Yuan](#), Y. Aloimonos.
International Conference on Robotics and Automation, [ICRA 2024](#). [Paper](#) [Code](#)
7. Gluing Neural Networks Symbolically Through Hyperdimensional Computing.
P. Sutor, [D. Yuan](#), D. Summer-Stay, C. Fermüller, Y. Aloimonos.
International Joint Conference on Neural Networks, [IJCNN 2022](#). [Paper](#) [Code](#)
8. Brain-Inspired Hyperdimensional Computing for Ultra-Efficient Edge AI.
H. Amrouch, M. Imani, X. Jiao, Y. Aloimonos, C. Fermüller, [D. Yuan](#), and other four authors.
2022 International Conference on Hardware/Software Codesign and System Synthesis. [Paper](#)

Publication — as a Research Assistant at HKUST.

As the second author of these papers, I processed geographic data > 100 GB, trained models using parallel computing, and generated all the geographic figures.

9. Global PM_{2.5} Prediction and Associated Mortality to 2100 under Different Climate Change Scenarios.
W. Chen, X. Lu, [D. Yuan](#), Y. Chen, Z. Li, Y. Huang, T. Fung, H. Sun, JCH. Fung
Environmental Science and Technology (I.F. 11.4) [Paper](#)
 10. Development of an Integrated Machine-Learning and Data Assimilation Framework for NO_x Emission Inversion.
Y. Chen, JCH. Fung, [D. Yuan](#), W. Chen, T. Fung, X. Lu
Science of Total Environment (I.F. 10.8) [Paper](#)
 11. Development of an LSTM Broadcasting Deep-learning Framework for Regional Air Pollution Forecast Improvement.
H. Sun, JCH. Fung, Y. Chen, Z. Li, [D. Yuan](#), W. Chen, X. Lu
Geoscientific Model Development Discussions (I.F. 6.9) [Paper](#)
 12. Estimation and Variation Analysis of Secondary Inorganic Aerosols Across the Greater Bay Area in 2005 and 2015.
Y. Chen, [D. Yuan](#), W. Chen, M. Hu, JCH. Fung, H. Sun, X. Lu
Chemosphere (I.F. 13.3) [Paper](#)
 13. Impacts of Urbanization and Long-Term Meteorological Variations on Global PM_{2.5} and its Associated Health Burden.
X. Lu, [D. Yuan](#), Y. Chen, JCH. Fung
Environmental Pollution (I.F. 8.9) [Paper](#)
 14. Estimations of Long-Term nss-SO₄²⁻ and NO₃⁻ Wet Depositions over East Asia by Use of Ensemble Machine-Learning Method.
X. Lu, [D. Yuan](#), JCH. Fung, W. Li, AKH. Lau
Environmental Science and Technology (I.F. 11.4) [Paper](#)
 15. Estimation and Spatiotemporal Analysis of NO₂ Pollution in East Asia During 2001–2016.
M. Hu, Y. Chen, [D. Yuan](#), R. Yu, X. Lu, JCH. Fung, W. Chen, Y. Huang, AKH. Lau
Journal of Geophysical Research: Atmospheres (I.F. 4.4) [Paper](#)
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Internship Experience

Applied Research Intern.

Jun. 3, 2024 – Aug. 23, 2024

Capital One, McLean VA.

- Same duties as my full-time Applied Researcher.

Data Scientist Intern.

Jun. 5, 2023 – Aug. 18, 2023

Liberty Mutual Insurance, Boston MA.

- Retrieve important activities from raw accident descriptions using **nltk** and **spacy**.
- Perform **clustering**, **association rule mining** on the important activities and visualize the analysis. Clients will know qualitatively and quantitatively how likely an activity will cause an accident and its average cost.
- Design a novel structured-pair representation of the activities, so that clients can easily understand the analysis. Design and implement an interface using **flask** so that clients can browse the analysis interactively.
- The entire pipeline streamlines the analysis of accident causes, which enables our company to enforce targeted safety measurement to different insurance purchasers.

Software Engineer Intern

Jun. 1, 2021 – Aug. 9, 2021

Invest Bots Limited, Hong Kong.

- Crawl and download annual reports (in pdf) using **bs4** and **requests**.
- Design and implement text mining algorithms to locate financial statements (~ 4 pages) from the annual reports (~ 200 pages). The localization accuracy reaches 97%.
- Retrieve tabular data from the financial statements with my novel images-of-table parsing algorithms. This module is used internally and also released at <https://table-reader.com>.
- The entire pipeline streamlines the downloading and parsing of financial statements, which enables companies to observe market features early.

Mathematical Foundation (Straight A/A+'s / Top 5%)

[my handwritten notes](#)

Fundamentals	Honor Calculus Honor Probability Honor Linear Algebra
Analysis	Real Analysis (up to measure theory) Complex Analysis (e.g. Residual Theorem) Theory of ODE (e.g. Poincaré-Bendixson Theorem)
Statistics	Statistical Inference, Regression Analysis, Stochastic Process
Computing	Computational Geometry, Ramsey Theory

Scholarships

- Research Fellowship awarded by Computer Science Department of UMD \$14,000 USD
- Dean's Fellowship awarded by UMD \$2,500 USD
- Scholarship awarded by HKUST \$30,000 HKD=\$3,846 USD (Top 5%)
- Scholarship awarded by Fok Yin Tung Research Institute, due to exceptional research productions. ¥60,000 RMB = \$9,500 USD
- The Epsilon Fund Award, due to exceptional performance in honor math courses. \$3,000 HKD=\$385 USD (Top 5%)
- Outstanding academic achievement in mathematics. \$30,000 HKD=\$3,846 USD (Top 1%)

Others

- Teaching Assistant for CMSC320 Introduction to Data Science for 3 semesters.
- Reviewer for CVPR, ICCV, ICLR, ICML, NIPS, AAAI, TPAMI, WACV.